CHEST

Interactive Pathways of Key Prognostic Factors in Severe Asthma

A Bayesian Network Comparison of Clinical Trials and Real-World Data

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BACKGROUND: The way in which risk predictors combine and contribute to severe asthma exacerbations may differ between clinical trials and real-world settings.

RESEARCH QUESTION: How do the interactive pathways of risk predictors leading to severe asthma exacerbations compare under clinical trials vs real-world settings?

STUDY DESIGN AND METHODS: The analysis involved 345 patients with severe asthma from the placebo arms of 2 international randomized controlled trials (RCTs), compared with 6,814 biologic-naïve patients from the International Severe Asthma Registry (ISAR). Sixteen key risk predictors, including demographics, biomarkers, lung function, health care use, exacerbation history, long-term oral corticosteroid use, asthma control, and nasal polyps, were covered. The outcome was the occurrence of severe asthma exacerbations over the 365 days after study enrollment. Bayesian networks (BNs), obtained from machine learning combined with expert knowledge, elucidated significant interplay processes of risk predictors that led to severe asthma exacerbations. External validation was performed in each cohort.

RESULTS: The RCTs revealed 44 significant arcs (ie, probabilistic interdependency) between 16 risk factors, whereas the ISAR showed 170. Despite this difference, the main downstream prediction pathways were consistent across both settings, with 2 key pathways: total serum IgE level influenced blood eosinophils to predict future severe exacerbations, and severe exacerbation history directly predicted future severe exacerbations. In external validation, RCT-BN generalized well to ISAR patients (area under the receiver operating characteristic curve, 0.68), whereas ISAR-BN underperformed in RCT patients (area under the receiver operating characteristic curve, 0.50), and ISAR-BN demonstrated better calibration.

INTERPRETATION: Our results show that the core pathways predicting severe asthma exacerbations were similar in both RCTs and real-world settings, with comparable predictive performance.

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KEY WORDS: evidence-based health care; machine learning; respiratory diseases; risk prediction

Severe asthma affects 5% to 10% of patients with asthma[1] but accounts for up to 60% of asthma-related health care costs,[2] imposing a disproportionately high health care and economic burden. Guideline-based management of severe asthma aims to reduce the ongoing risk of severe exacerbations, defined by the 2014 criteria of the European Respiratory Society and American Thoracic Society[3] as worsening asthma requiring ≥ 3 days of oral corticosteroids (OCSs). Accurate identification of high-risk patients would enhance disease management and enable personalized risk reduction and treatment strategies, empowering patients through shared clinical decision-making.[4,5]

Both randomized controlled trials (RCTs) and real-world data (RWD) have been used to derive asthma prediction tools. RCTs have been used to identify risk predictors, estimate predictive capacity of treatment effect, and derive risk prediction models in asthma, such as the OxfoRd Asthma attaCk risk ScaLE.[6,7] Meanwhile, advanced machine learning (ML) risk prediction models have been developed from RWD such as electronic health records, real-time self-collected data for home monitoring, and subjective composite scales such as the Asthma Impairment and Risk Questionnaire tool.[8-11]

RCTs for severe asthma typically exhibit high internal validity due to stringent participant selection criteria, but limited external validity due to small sample size and restricted clinical environment. In contrast, RWD studies encompass a broader demographic, providing

ABBREVIATIONS: ACQ-5 = Asthma Control Questionnaire-5; AUC = area under the receiver operating characteristic curve; BEC = blood eosinophil count; BN = Bayesian network; DREAM = Dose Ranging Efficacy and Safety With Mepolizumab; ED = emergency department; FENO = fractional exhaled nitric oxide; ISAR = International Severe Asthma Registry; MENSA = Mepolizumab as Adjunctive Therapy in Patients with Severe Asthma; ML = machine learning; OCS = oral corticosteroid; RCT = randomized controlled trial; RWD = real-world data; SAE = severe asthma exacerbation

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Take-Home Points

Research Question: How do the interactive pathways of risk predictors for severe asthma exacerbations differ between randomized controlled trials and real-world settings?

Results: The main downstream prediction pathways were consistent across both settings, with 2 key pathways: (1) total serum IgE level influenced blood eosinophils to predict future severe exacerbations, and (2) history of severe exacerbation directly predicted future severe exacerbations.

Interpretation: The essential pathways predicting severe asthma exacerbations were similar in both randomized controlled trials and real-world settings, with comparable predictive performance.

insights into treatment effectiveness in everyday clinical settings, thus enhancing generalizability and complementing RCT findings.[12] It remains unclear whether the composition and interaction of risk predictors differ between RCTs and RWD, which could impact the applicability of RCT-derived prediction rules into real-world settings. Additionally, the aforementioned models did not inform the interactive pathways between risk predictors and outcome, which could hamper knowledge translation and physician trust in clinical use.[13] To better understand the intimate pathways of risk factors under RCTs and RWD settings, a Bayesian network (BN)[14] methodology could be a solution. A BN is a probabilistic graph that represents dependence (cause-and-effect relationships) between predictors and can be used in medical diagnostics under uncertainty.[14]

Applying the novel BN approach, this study compared the associations and interactions among core predictors leading to severe asthma exacerbations (SAEs) between a pooled cohort of 2 multisetting RCTs of biologic therapy,[15,16] and the largest international registry cohort of severe asthma across 20 countries.[17]

Study Design and Methods

Data Source

We retrieved data from 2 individual-level RCTs and RWD focusing on adults who had experienced SAEs. The first RCT, Dose Ranging Efficacy and Safety With Mepolizumab in Severe Asthma (DREAM) (November 2009 to December 2011)[16] enrolled 616 patients across 81 centers in 13 countries. The second RCT, Mepolizumab as Adjunctive Therapy in Patients with Severe Asthma (MENSA) (October 2012 to January 2014),[15] involved 570 patients and 16 countries. Details of the inclusion-exclusion criteria are given in e-Appendix 1.

For RWD, we extracted data from the International Severe Asthma Registry (ISAR) (2017-2021, https://isaregistries.org/),[17] comprising 12,458 patients from 19 countries diagnosed with severe asthma receiving Global Initiative for Asthma step 5 treatment or with severe asthma remaining uncontrolled at Global Initiative for Asthma step 4. The ISAR data included demographics, health behaviors, asthma specifics, lung function, inflammation biomarkers, comorbidities, health care utilization, and asthma exacerbation outcomes.

The index date of a patient was defined as the date of the baseline visit to the asthma clinic to join the study.

Study Population

We analyzed pooled individual-level data from DREAM and MENSA, focusing on patients from the placebo arm who had $\leq$ 1 year of follow-up, for a total of 345 patients (Fig 1A). We included participants $>$ 12 years of age to avoid any further exclusions, given the modest sample size relative to the ISAR. For the ISAR, we included patients with severe asthma $\geq$ 18 years of age who never received biologic therapy at baseline, for a total of 7,771 patients. We followed an intention-to-treat design for the ISAR, for consistency with the RCT population. After inclusion-exclusion criteria (Fig 1B) and excluding individuals with $<$ 300 days of follow-up, and capping at 365 days, 6,814 patients (55% of the ISAR) were included. Participants with $<$ 300 days of follow-up were excluded to ensure adequate outcome measurement because follow-up time was not used as a predictor or offset.

Outcome Variable

The outcome, frequency of future SAEs, was defined as an asthma attack necessitating $\geq$ 3 days of OCS use, and/or resulting in asthma-related hospitalization, or emergency department (ED) visits, aligning with the American Thoracic Society/European Respiratory Society Task Force definition.[3] The measurement was censored at 365 days, and categorized into 0, 1, and $\geq$ 2 exacerbations.

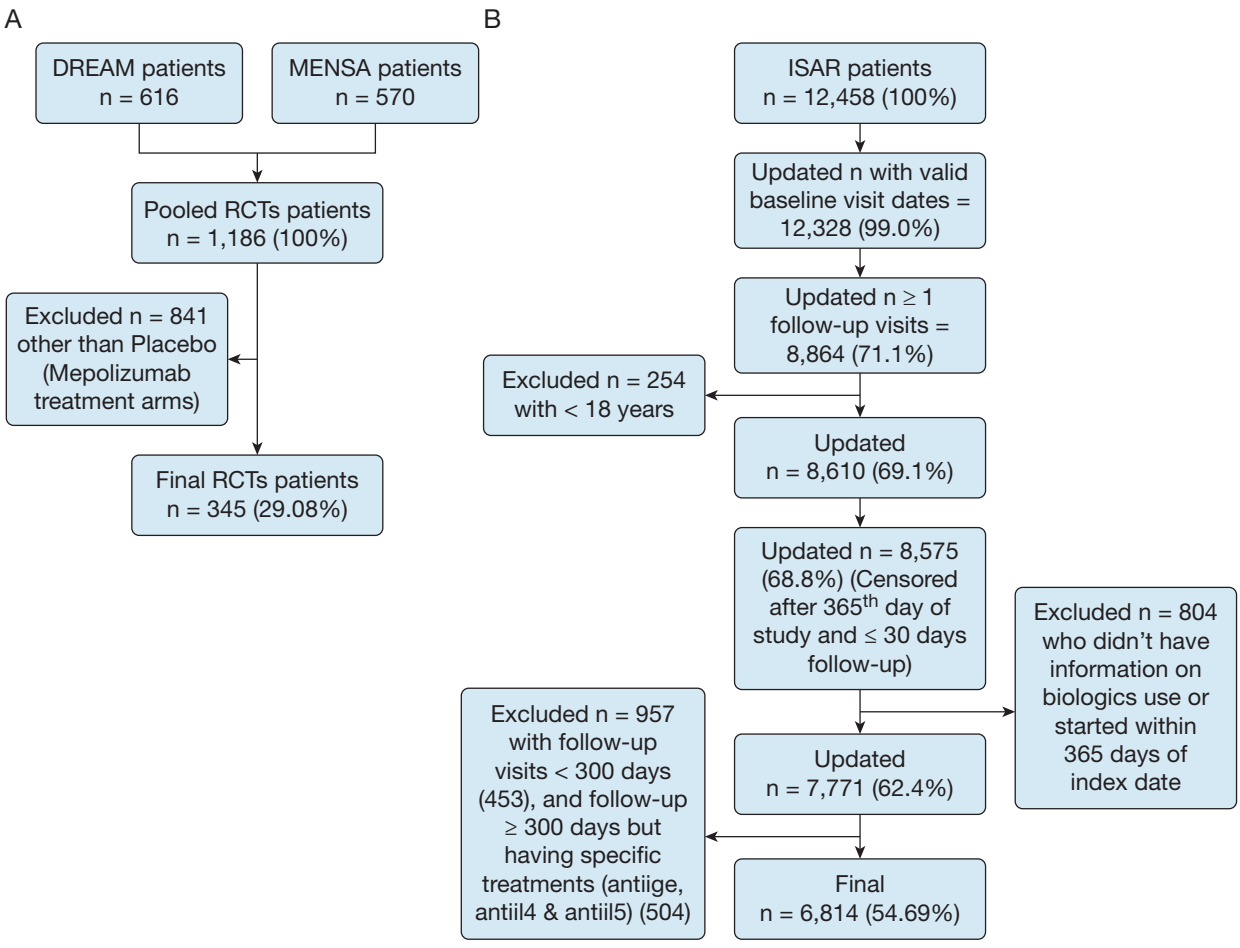


Figure 1 – *A, B, Patient selection flowchart for the RCTs (A) and ISAR (B). DREAM = Dose Ranging Efficacy and Safety With Mepolizumab; ISAR = International Severe Asthma Registry; MENSA = Mepolizumab as Adjunctive Therapy in Patients with Severe Asthma; RCT = randomized controlled trial.*

Core Predictors

Following our recent study,[6] we included prognostic predictors which were common in the 2 RCTs and the ISAR. A total of 16 key prognostic predictors including age, sex (male or female), race/ethnicity (RCTs: Hispanic or Latino, not Hispanic or Latino; ISAR: White, Black, multiracial, North East Asian, South East Asian, others [eg, Middle Eastern, South Asian, or unclear categorization]), BMI (underweight: < 18.5 kg/m^2, normal: 18.5-24.9 kg/m^2, overweight: 25-29.9 kg/m^2, and obese: > 30 kg/m^2), smoking history (yes or no), blood eosinophil count (BEC), % predicted prebronchodilator FEV_1,[18] prebronchodilator FEV_1/FVC ratio, % FEV_1 reversibility (an increase of at least 12%, or 200 mL in FEV_1 after taking bronchodilators), total serum IgE level, Asthma Control Questionnaire-5 (ACQ-5) scores[19] (ACQ-5 < 0.75 indicating well-controlled asthma, ACQ-5 of 0.75-1.5 indicating partially controlled asthma, and ACQ-5 > 1.5 indicating uncontrolled asthma), presence of nasal polyps (yes or no), long-term use of OCSs (yes or no), past 12-month asthma-related hospitalization, ED visits, and SAEs. Of note, fractional exhaled nitric oxide (FENO) was not available in MENSA, and was excluded from our analysis.

We performed the significance test with Bonferroni correction for predictors based on Wilcoxon rank-sum tests for continuous predictors and χ^2 tests for categorical predictors.

Statistical Analysis

Data Processing: Outliers were independently evaluated, and extreme values were recoded as missing. Variables with moderate missingness (serum IgE, 41.2%; others, $< 30\%$) (e-Table 1) were imputed using the MissForest[20] algorithm, a nonparametric random forest approach which yields robust performance across missingness mechanisms and experimental settings for $\leq$

50% missingness.[21] For better clinical interpretability and minimal information loss, we discretized continuous predictors based on clinically relevant cutoffs, and Hartemink criteria.[22] To address unbalanced class distributions in both data sets, we used the Adaptive Synthetic Sampling Algorithm[23] on discretized training data, with sampling weighted by learning difficulty of the specific class. This approach sought to address data distribution gaps between sources by improving city class performance and reducing majority class bias.

BN Learning: To explore interactions among prognostic risk predictors, we applied the constraint-based algorithm semi-interleaved HITON-PC[24] to map out the local network of each predictor, identifying potential influencers (parents), and those that could be influenced (children). This was complemented by the Max-Min Parents and Children[25] algorithm to establish the direction of connecting arcs.

A hybrid approach combining expert knowledge with ML algorithms was used to select important arcs and parametrize the final model. We used 5 ML algorithms[25] for BN learning: Hill Climbing, Tabu Search, Peter-Clark, Grow Shrink, and 2-phase Restricted Maximization. To mitigate overfitting, each algorithm underwent 1,000 rounds of bootstrapping with a 0.75 threshold for arc inclusion—a conservative criterion for identifying reliable relationships.[14] The final BN included arcs identified by ≥ 2 algorithms (to balance robustness and sensitivity in network structure learning), refined by expert insights to address clinical relevance and direction. Using multiple algorithms ensured that only consistently supported arcs were retained, reducing the likelihood of spurious edges.

We adjusted for country variation in SAE frequency using a linear mixed-effects model to account for unexplained country-level variability. For pooled RCT data, predictors identified as children of the country predictor were included as fixed effects, whereas country itself was included as a random effect. For the ISAR, fixed effects were expanded to include both children of the country variable and the biologic accessibility score,[26] adjusting for unobserved heterogeneity among nonbiologic patients, with country remaining a random effect. The estimated country-specific random effects were subtracted from the observed outcome to generate a residualized exacerbation frequency, which was then discretized (0, 1, or ≥ 2) and used as the target node. This approach allowed the network to capture patient-level relationships while accounting for systematic differences across countries.

Model Validation: The BN developed using RCT data (RCT-BN) was validated externally on the original and synthetic ISAR data. BN developed using ISAR data (ISAR-BN) was validated externally on the original and synthetic pooled RCTs data set.

The model was evaluated using area under the receiver operating characteristic curve (AUC), accuracy, precision, recall, and F1 score. AUC values near 1 indicate good classwise discrimination. Accuracy, the ratio of correctly predicted observations to total observations, is best suited for balanced classes. For imbalanced classes, precision (true positives to all positives) is crucial when false positives are costly, and recall (true positives to actual positives), is vital when false negatives are costly. The F1 score, the harmonic mean of precision and recall, balances these metrics.

Calibration under pooled RCTs and the ISAR were plotted using 70:30 train test data methods (ie, we trained models on 70% of observations and predicted for the next 30%).[27] Calibration intercepts of < 0, > 0, and 0 indicate underestimation, overestimation, and perfect calibration, respectively. Calibration slopes of 1, < 1, and > 1 indicate perfect, overfitted, and underfitted calibrations, respectively.

All analyses were conducted using R software (R Foundation for Statistical Computing; https://www.r-project.org/), and codes can be obtained on request. This study has obtained ethics approval from the institutional review board of National University of Singapore (NUS-IRB-2021-877) and the Anonymized Data Ethics & Protocol Transparency Committee (ADEPT1924).

Results

Figure 1 presents the patient selection flowcharts of the RCTs and ISAR cohorts. Table 1 presents patient characteristics and their differences between the RCTs and ISAR cohorts. In the RCTs, the mean age ± SD was 47.9 ± 13.1 years, with 203 female patients (58.8%). The mean rate of SAE ± SD was 3.6 ± 3.3 in the past year, and 1.4 ± 1.7 in the follow-up year. In the ISAR, the mean age ± SD was 55.2 ± 15.1 years, with 4,172 female patients (61.2%). The mean rate of SAE ± SD was 0.5 ± 1.2 in the past year, and 0.2 ± 0.8 in the follow-up year. According to the differential distributions of core prognostic factors

TABLE 1] Patient Characteristics in the RCTs and ISAR

Predictor	RCTs	ISAR	*P* Value	Adjusted *P* Value (Bonferroni)
Age, y	47.9 [13.1]	55.2 [15.1]	< .001	< .001
Female	203 (58.8)	4,172 (61.2)	.4	> .99
Smoking history, yes	58 (16.8)	2357 (34.6)	< .001	< .001
BMI, kg/m^2	28.1 [5.8]	29.1 [6.5]	.010	.070
Nasal polyps	49 (14.2)	2,120 (31.1)	< .001	< .001
% predicted prebronchodilator FEV_1	60.3 [16.9]	73.6 [20.0]	< .001	< .001
FEV_1 reversibility, %	21.1 [23.9]	5.5 [9.1]	< .001	< .001
Prebronchodilator FEV_1/FVC ratio	0.6 [0.1]	0.7 [0.1]	< .001	< .001
Asthma control, well	21 (6.1)	926 (13.6)	< .001	< .001
Blood eosinophil count, × 10^9/L	450.1 [402.5]	426.4 [326.2]	.5	> .99
Serum IgE, Units/mL	473.2 [1,194.9]	406.7 [884.9]	.010	.2
Use of long-term oral corticosteroids, yes	146 (0.4)	514 (7.5)	< .001	< .001
No. of severe exacerbations in the previous year	3.6 [3.3]	0.5 [1.2]	< .001	< .001
No. of severe exacerbations required hospitalization in previous year	0.3 [0.6]	0.1 [0.36]	< .001	< .001
No. of severe exacerbations required emergency department visits in previous year	0.5 [2.8]	0.1 [0.6]	.1	> .99

Data are presented as No. (%), mean [SD], or as otherwise indicated. ISAR = International Severe Asthma Registry; RCT = randomized controlled trial.

and health care utilizations, the RCT cohort had more patients with severe asthma than the ISAR, both before and after Bonferroni adjustment (Table 1). e-Table 2 provides details on the registry-specific asthma diagnosis and severe asthma categorization. e-Table 3 presents additional information of the RCT and ISAR cohorts. e-Table 4 represents the categories of risk predictors after categorizing the continuous predictors under RCTs and ISAR.

Figure 2 illustrates the local neighborhood of predictor associations identified by the semi-interleaved HITON-PC algorithm in RCTs (Fig 2A) and ISAR (Fig 2B) settings. In the RCTs, 44 arcs were identified in the neighborhood connections of 17 predictors. The nodes with the highest connectivity included total ED visits (4 arcs), % predicted prebronchodilator FEV_1 (3 arcs), total serum IgE (3 arcs), OCS use (3 arcs), % FEV_1 reversibility (3 arcs), history of severe exacerbations (3 arcs), and smoking history (3 arcs). On the downstream pathways, 6 predictors, including BEC, history of SAEs, asthma control, OCS use, smoking history, and total ED visits, were directly connected to the outcome (ie, 12-month SAEs).

In the ISAR, 170 arcs were identified: the most connected arcs involved history of SAEs (14 arcs), % predicted prebronchodilator FEV_1 (14 arcs), sex (14 arcs), BMI (14 arcs), age (13 arcs), BEC (12 arcs), prebronchodilator FEV_1/FVC ratio (11 arcs), % FEV_1 reversibility (10 arcs), IgE level (10 arcs), smoking history (9 arcs), and presence of nasal polyps (8 arcs). On the downstream, 15 variables, including age, sex, ethnicity, IgE level, BEC, BMI, OCS use, % predicted prebronchodilator FEV_1, prebronchodilator FEV_1/FVC ratio, FEV_1 reversibility, history of SAEs, total ED visits, nasal polyps, total hospital visits, and asthma control, were directly connected to the outcome.

Figure 3 shows BNs learned from ML algorithms integrating expert knowledge under RCTs and ISAR settings. The bold lines between predictors showed that these predictors were in the direct prediction pathways; the dotted lines indicated these predictors were consequences of others that do not lead to the outcome. Figure 4 depicts the final BN after removing noncore predictors.

RCT Settings

Two separate downstream pathways were identified: (1) IgE influenced BEC to influence future SAEs, and (2) history of SAEs independently influenced future SAEs (Figs 3A, 4A). On the upper stream, smoking

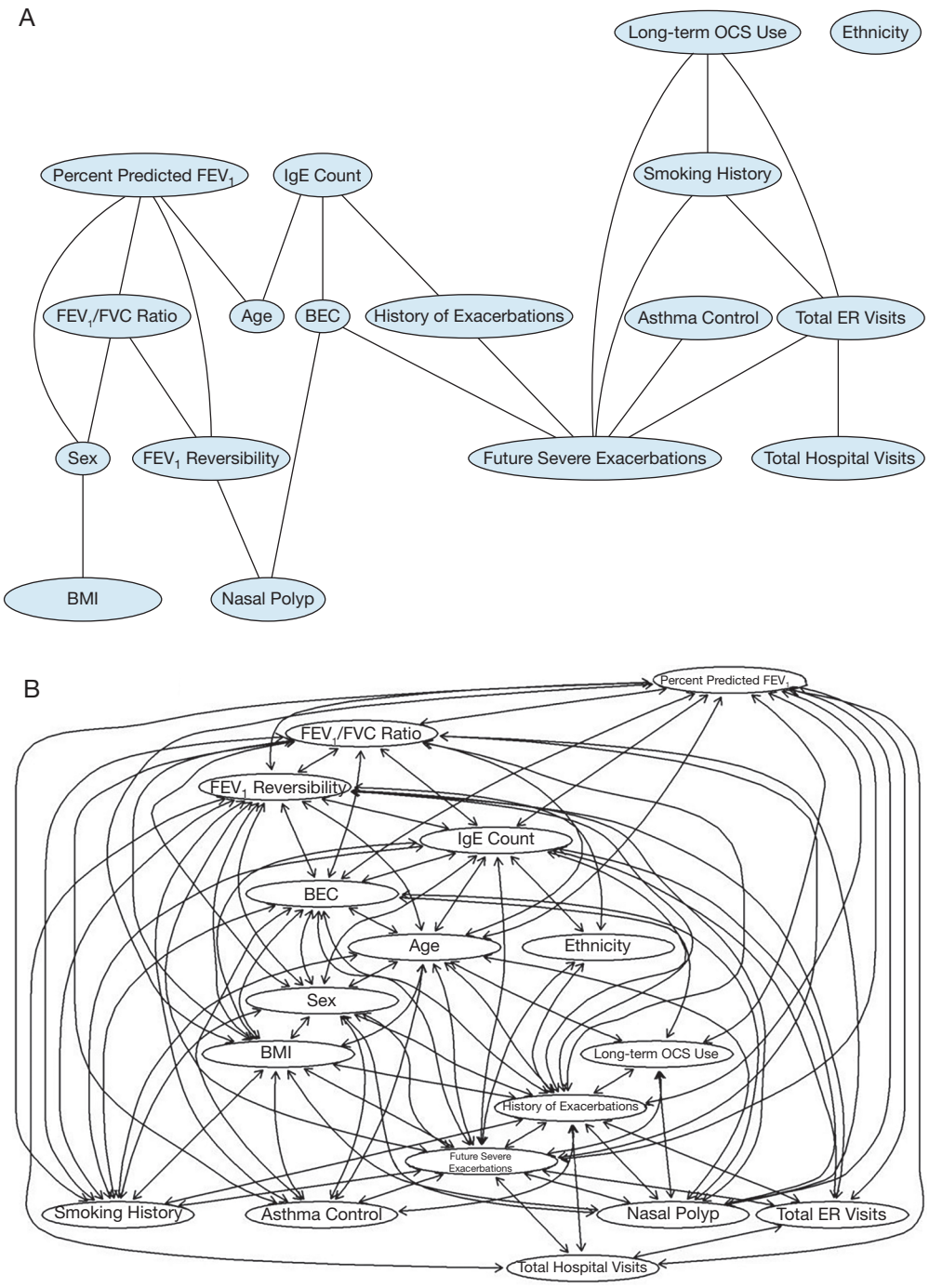


Figure 2 – *A, B, Structure plot showing parents and children among predictors for the randomized controlled trials (A) and International Severe Asthma Registry (B). BEC = blood eosinophil count; ER = emergency room; OCS = oral corticosteroid.*

history affected OCS use, which in turn influenced the history of severe asthma. Age, ethnicity, BMI, asthma control, and lung function (influenced by sex) did not significantly influence any predictors or outcome. Past 12-month hospitalizations and nasal polyps were the end points of other predictors' interactions.

ISAR Setting

Two separate downstream pathways were identified, which were identical in the RCTs (Figs 3B, 4B). However, the upstream predictor interplays were more complex than those in the RCTs: (1) age and sex influenced IgE to affect BEC; and (2) age and sex influenced lung function, and then lung function

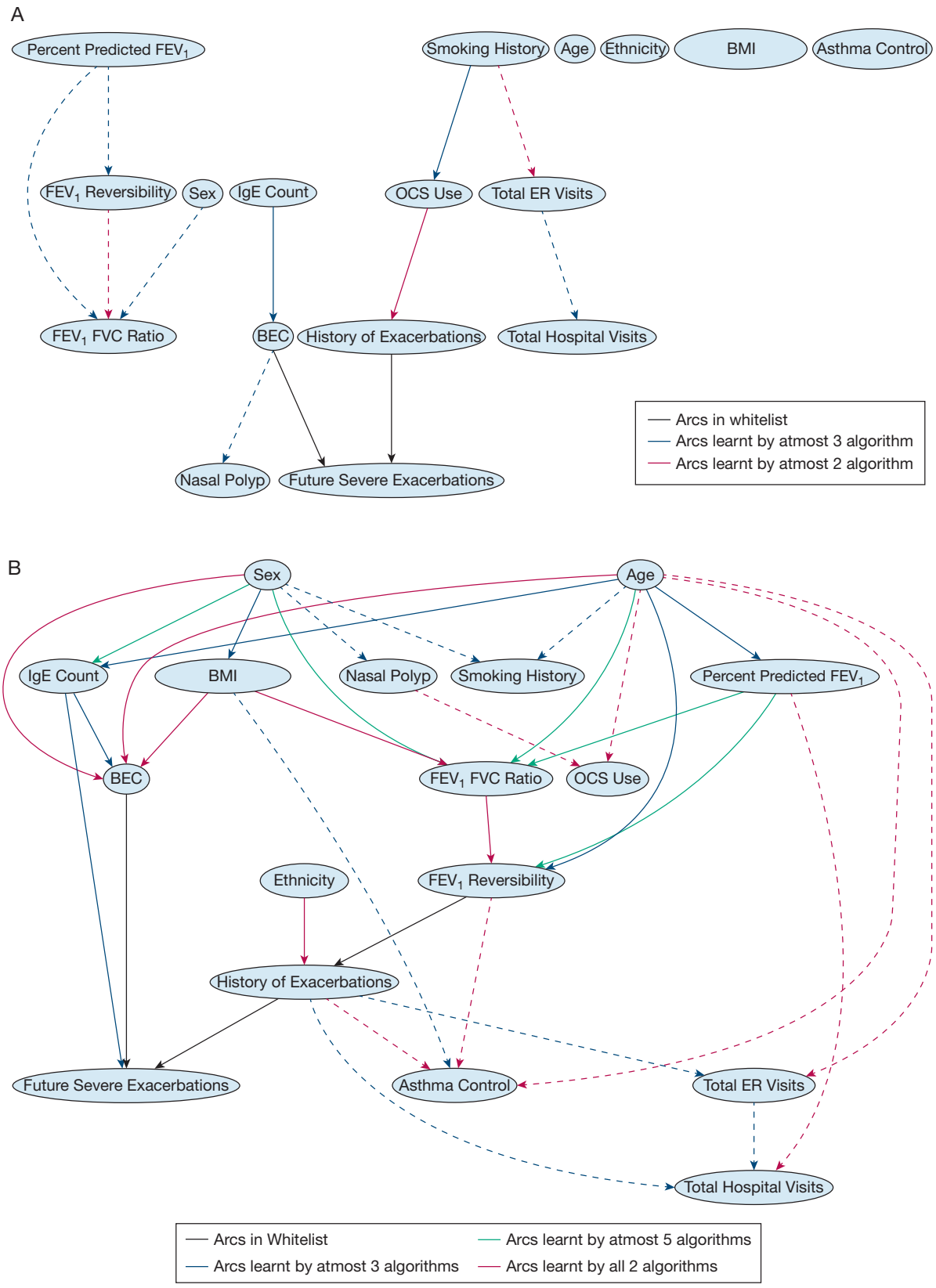


Figure 3 – *A, B, Bayesian network from combined machine learning algorithms for the randomized controlled trials (A) and International Severe Asthma Registry (B). Boldface lines indicate the direct predictors of the outcome; dotted lines indicate predictors without direct impact. BEC = blood eosinophil count; ER = emergency room; OCS = oral corticosteroid.*

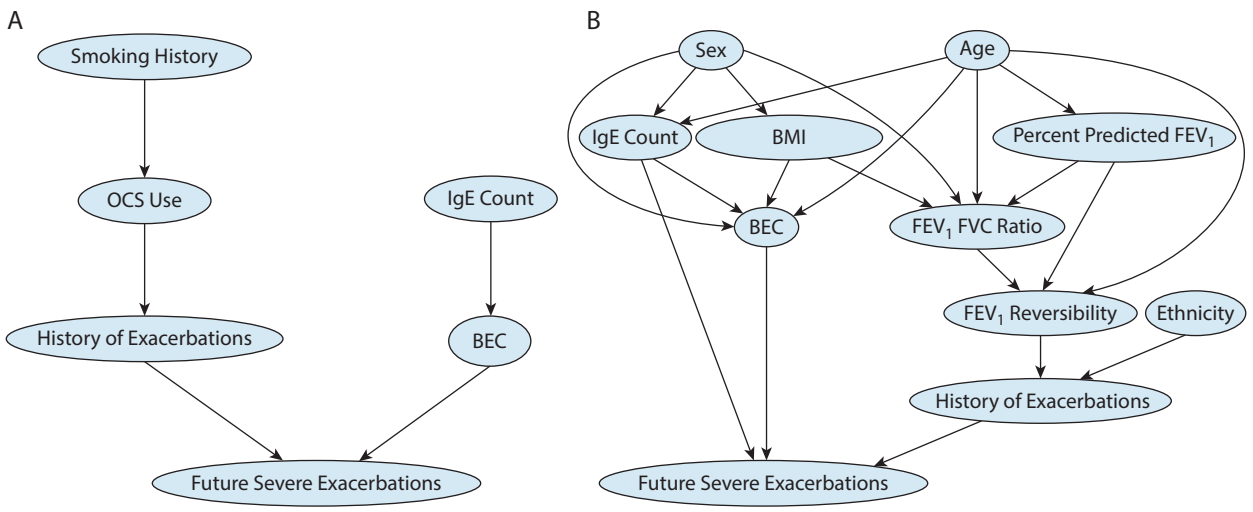


Figure 4 – *A, B, Final Bayesian network learned under the randomized controlled trials (A) and International Severe Asthma Registry (B) settings. BEC = blood eosinophil count; OCS = oral corticosteroid.*

influenced history of SAE together with ethnicity. Past 12-month hospitalizations, OCS use, asthma control, and smoking history were end points of other predictors' interactions.

Table 2 presents the external validity of the final BNs, which were respectively derived from the RCTs and ISAR on the synthetic sample. The RCT-BN yielded good and consistent discriminative capacity when externally validated in both the synthetic and original RWD samples (AUC, 0.67 and 0.68, respectively). However, the ISAR-BN demonstrated adequate performance in the synthetic RCT sample but poorer performance in the original RCT sample (AUC, 0.62 and 0.50, respectively), suggesting reduced transferability of RWD-learned relationships to the more restricted RCT setting.

Meanwhile, the ISAR-BN yielded better model calibration than the RCT-BN. In the RCT-BN, calibration intercepts for predicting ≥1 and ≥ 2 SAEs were 0.01 and −0.55, and calibration slopes were 0.84 and 0.65, respectively (Fig 5). In the ISAR-BN, calibration intercepts for ≥ 1 and ≥ 2 SAEs were 0.02 and −0.002, and calibration slopes were 0.95 and 1.02, respectively (Fig 6).

Discussion

This study extends on our recent work[28] by examining the reliability and complementarity of SAE prediction pathways across evidence sources. Our findings demonstrated stable core mechanistic relationships across RCTs and RWD, whereas the latter provided enriched upstream contexts. This study offers insights

TABLE 2] External Validation[a]

	RCT-Derived BN in the ISAR Cohort		ISAR-Derived BN in the RCT Cohort	
Performance Metric	Adaptive Synthetic Cases	Original Cases	Adaptive Synthetic Cases	Original Cases
AUC	0.67	0.68	0.62	0.50
Specificity	0.74	0.76	0.71	0.67
Precision	0.50	0.49	0.41	0.83
Recall	0.50	0.50	0.42	0.33
F1 score	0.49	0.47	0.40	0.91
Accuracy	0.62	0.63	0.56	0.50

The ranges for these metrics vary from 0 to 1, where higher values indicate better performance. Accuracy = proportion of correct predictions made by the model among all predictions; AUC = area under the receiver operating characteristic curve, which plots the true-positive rate against the false-positive rate; BN = Bayesian network; F1 score = harmonic mean of precision and recall and provides a balance between precision and recall; ISAR = International Severe Asthma Registry; Precision = proportion of true positives among all positive predictions made by the model; Recall = proportion of true positives correctly identified among all actual positive instances; RCT = randomized controlled trial; Specificity = proportion of true negatives correctly identified by the model.

[a]External validation tested RCT-BN in the ISAR population and ISAR-BN in the RCT population.

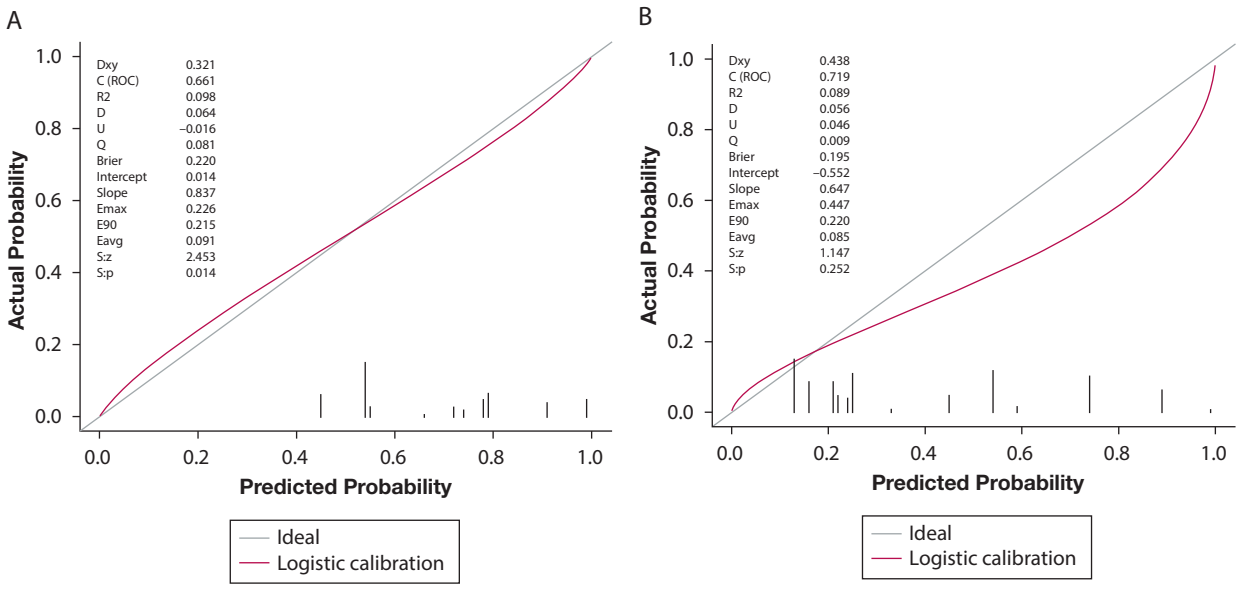


Figure 5 – *A, B, Calibration plot of predicting 1 (A) and 2 (B) exacerbations in the randomized controlled trials. C = C-statistic, concordance index (equivalent to area under the receiver operating characteristic curve [AUC]); D = discrimination index; Dxy = somers' D rank correlation; E90 = 90th percentile of the absolute difference between observed and predicted probabilities; Eavg = average absolute calibration error; Emax = maximum absolute difference between observed and predicted probabilities; Q = overall quality; R2 = coefficient of determination; S:p = significance (*P *value); S:z = significance (*z *statistic); U = unreliability index.*

to bridge the gap between the controlled conditions of clinical trials and the complex realities of real-world practice, which may lead to more accurate and generalizable risk prediction models in severe asthma.

The downstream prediction pathways were similar across RCTs and RWD settings. The first pathway was characterized by serum IgE influencing BEC, and the latter directly influenced future SAEs. In the second

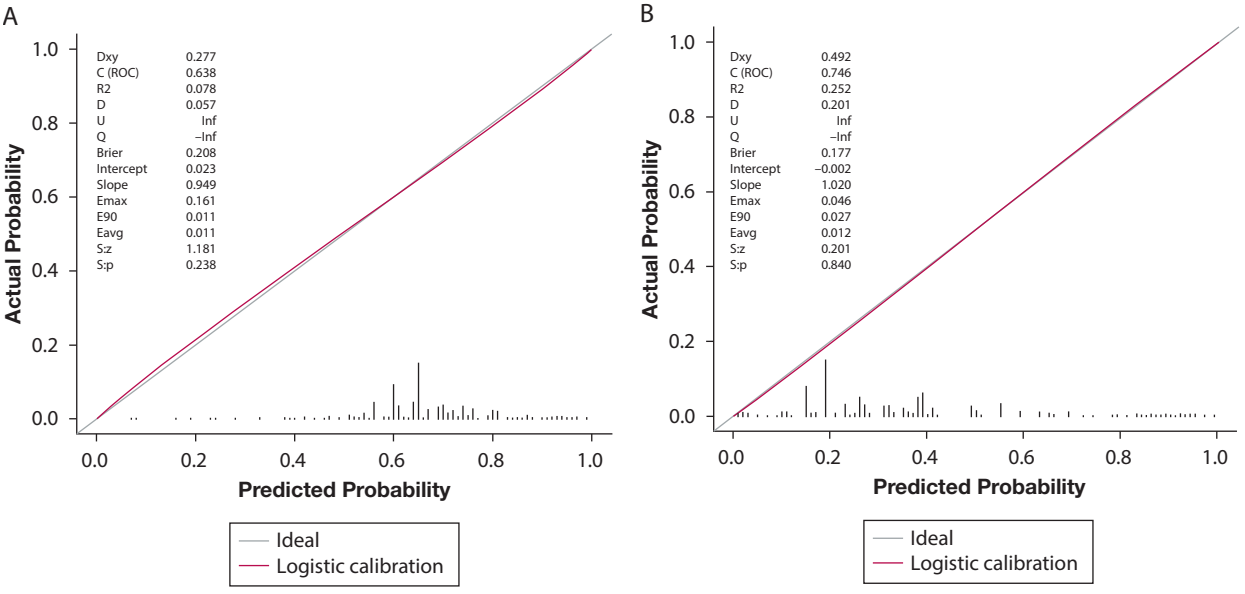


Figure 6 – *A, B, Calibration plot of predicting 1 (A) and 2 (B) exacerbations in the International Severe Asthma Registry. C = C-statistic, concordance index (equivalent to area under the ROC curve [AUC]); D = discrimination slope; Dxy = somers' D rank correlation; E90 = 90th percentile of the absolute difference between observed and predicted probabilities; Eavg = average absolute calibration error; Emax = maximum absolute difference between observed and predicted probabilities; Q = overall quality; R2 = coefficient of determination; S:p = significance (*P *value); S:z = significance (*z *statistic); U = unreliability index.*

pathway, historical exacerbations alone directly influenced future SAEs.[29] Interpretability regarding the indirect role of serum IgE merits caution because the current analysis did not include FENO, another key biomarker of T2 inflammation whose predictive capacity of future SAEs was well demonstrated.[30] The downstream pathways may change with potential inclusion of FENO, but FENO was unavailable in the MENSA trial.[15,30] In an ISAR-based preliminary analysis which included FENO, serum IgE would no longer be involved in the essential prediction pathway of SAEs, supporting findings that serum IgE alone was not predictive of exacerbations.[31] Rather, it likely served as an indirect biomarker of IL4/13[32] instead of being necessarily causative in the current BN.

Although RWD are prone to unmeasured confounding, their impact—such as variations in model structure and performance—was primarily observed in upstream prediction pathways. Key biomarkers and a history of SAE proved robust in identifying high-risk patients across data environments, underscoring their predictive utility. Meanwhile, ISAR-BN incorporated more key predictors in the upstream pathways. Unlike the structured environment of RCTs, RWD are characterized by diverse patient populations, missing data, and the multifaceted nature of real-world patient care. Besides FENO, crucial risk factors such as comorbidities, atopy status, socioeconomic status, environmental exposures, and medication use were also unavailable in MENSA and DREAM data, representing a limitation in RCT-BN. In this aspect, RWD offer an opportunity to fill this data gap, given the demonstrated comparability in prediction pathways. Future studies should incorporate decision-analytical metrics to assess the incremental value of BN models, and explore penalized BN or reduced predictor sets to further mitigate overfitting and enhance model stability.

Although the historical exacerbation rate was significantly higher in RCTs than in RWD (3.6 vs 0.5 events in placebo/biologic-naive patients), which likely reflects stricter eligibility criteria in trial settings, the predictive performance of the RCT-BN and ISAR-BN was comparable in the balanced synthetic samples. The RCT-BN demonstrated better discriminative capacity, whereas the ISAR-BN showed higher calibration. The commendable balance between precision and recall of RCT-BN suggests effective identification of positive cases while minimizing false positives and false negatives. However, to be reliably informed about actual risks, calibration may be more relevant to clinical decision-making. ISAR-BN demonstrated near-ideal calibration for predicting both any and frequent exacerbations, whereas the RCT-BN's negative intercepts indicated overestimation, likely due to the greater severity and homogeneity of the RCTs populations. This discrepancy which may limit the effectiveness of RCT-BN in identifying patients with the highest SAE risk. Nonetheless, RCT-BN demonstrated stable external validity in the original ISAR sample, whereas RWD-BN showed limited generalizability to the original RCT sample. This likely reflects the inherent differences between data environments: RCTs represent more homogeneous and controlled populations, allowing transportability of the learned relationships to real-world contexts, whereas RWD-BN may involve overfitting to context-specific patterns, which reduces generalizability to the more restrictive RCT environment.

This study has several limitations. First, RCT findings should be interpreted with caution due to the limited sample size and strict eligibility criteria, which may reduce BN learning performance.[33] The event per variable analysis in RCTs showed overfitting risk for outcome category 1 (event per variable, 5.6) and suboptimal values that fell below the recommended threshold of 10 for categories 0 (8.5) and $\geq$ 2 (7.4).[33] Second, RCT enrollees often demonstrate better lung function than patients in observational studies.[34] Third, the RCT placebo group included patients eligible for biologics, whereas a considerable portion of nonbiologic patients in the ISAR did not qualify for biologics; these inherent differences were already reflected in the baseline and follow-up frequencies of exacerbations. Fourth, the use of synthetic cases to balance outcome classes may distort real-world incidence probabilities. Fifth, the inherent biases and limitations of RWD, including confounding and missing data, have been discussed in the ISAR literature.[35] Sixth, variation in racial/ethnic categorization and country composition across settings may limit the comparability of findings. Seventh, including ISAR countries beyond those in the pooled RCTs may introduce bias due to geographic differences. This was mitigated by rigorous mixed-effects adjustment for country variation in RWD and RCTs. Moreover, country variation under standardized RCT protocols are typically not of a significant magnitude that would reverse the direction of effect estimations.[36] Eighth, continuous variables were discretized using clinically relevant cutoffs and the Hartemink criterion[22] to ensure interpretability and

minimize information loss. Finally, data collection methods, predictor measurements, and severe asthma definition may differ across ISAR participating centers. Global and temporal variations in SAE risk exist, influenced by health care systems, policies, and environmental factors.[37]

Interpretation

Despite complexities of real-world settings, the downstream prediction pathways for future SAEs were consistent across RCTs and the ISAR. In the upstream pathways, the ISAR-BN captured more complex interactions involving additional risk factors. Model validation showed comparable performance between the RCT-derived and ISAR-derived BNs. These findings demonstrated that key predictors remain consistent across RCTs and RWD, despite differences in data collection and patient populations, underscoring the potential of RWD to complement RCTs data in predictive modeling, particularly when RCT data are limited.

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